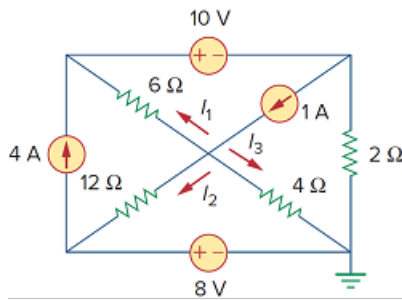


Problem

In the circuit of Fig. 3.100, solve for I_1 , I_2 , and I_3 .

Figure 3.100:



Step-by-step solution

Step 1 of 11

Refer to figure 3.100 given in the textbook.

Redraw the circuit with the loop currents as shown in figure 1.

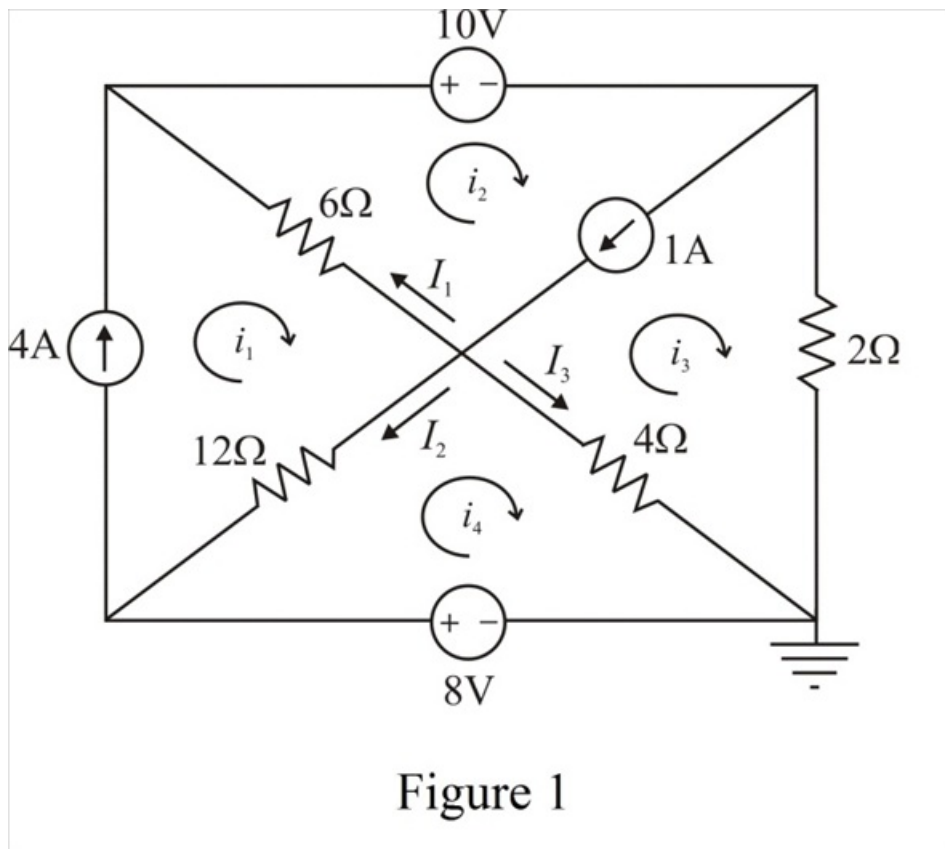


Figure 1

Step 2 of 11

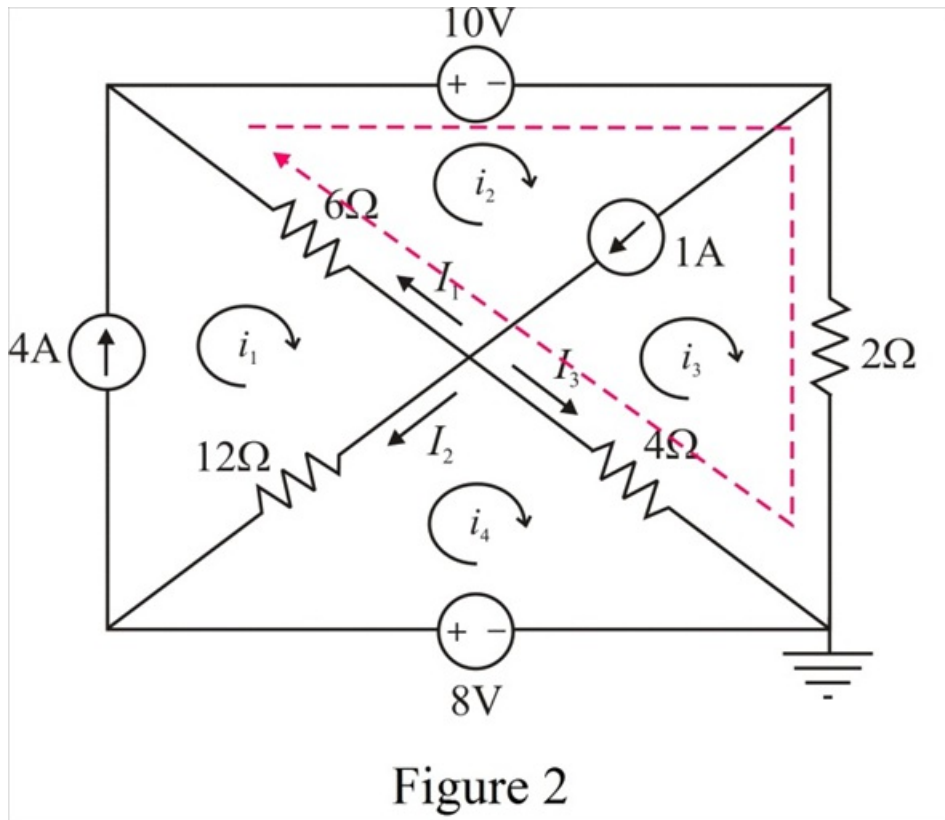
Consider mesh 1.

The value of mesh current i_1 is,

$$i_1 = 4 \text{ A}$$

Observe the super-mesh results as the two meshes have a dependent or independent current source in common. An 1 A independent current source exists between mesh 2 and mesh 3 as shown in figure 2.

Therefore, represent the super-mesh as shown in figure 2.



Apply Kirchhoff's voltage law to the super-mesh of the circuit shown in figure 2.

$$-10 - 2i_3 - 4(i_3 - i_4) - 6(i_2 - i_1) = 0$$

$$-6i_3 + 4i_4 - 6i_2 + 6i_1 = 10$$

$$-3i_3 + 2i_4 - 3i_2 + 3i_1 = 5$$

Substitute 4 for i_1 .

$$-3i_3 + 2i_4 - 3i_2 + 3(4) = 5$$

$$-3i_3 + 2i_4 - 3i_2 + 12 = 5$$

$$3i_2 + 3i_3 - 2i_4 = 7 \dots\dots (1)$$

Apply Kirchhoff's current law to the super-mesh shown in figure 2.

$$i_2 - i_3 = 1$$

$$i_2 = 1 + i_3 \dots\dots (2)$$

Apply Mesh analysis to mesh 4 of the circuit shown in figure 1.

$$8 - 12(i_4 - i_1) - 4(i_4 - i_3) = 0$$

$$12i_1 + 4i_3 - 16i_4 = -8$$

Substitute 4 for i_1 .

$$12(4) + 4i_3 - 16i_4 = -8$$

$$48 + 4i_3 - 16i_4 = -8$$

$$16i_4 - 4i_3 = 56$$

$$4i_4 - i_3 = 14$$

$$i_4 = \frac{i_3 + 14}{4} \dots\dots (3)$$

Step 6 of 11

Recall equation (1).

$$3i_2 + 3i_3 - 2i_4 = 7$$

Consider equation (2) and equation (3). Substitute $(1 + i_3)$ for i_2 , and $\left(\frac{i_3 + 14}{4}\right)$ for i_4 .

$$3(1 + i_3) + 3i_3 - 2\left(\frac{i_3 + 14}{4}\right) = 7$$

$$3 + 3i_3 + 3i_3 - \frac{i_3}{2} - 7 = 7$$

$$3 + 6i_3 - \frac{i_3}{2} - 7 = 7$$

$$6 + 12i_3 - i_3 - 14 = 14$$

Simplify further,

$$11i_3 = 14 + 14 - 6$$

$$11i_3 = 22$$

$$i_3 = 2 \text{ A}$$

Step 7 of 11

Recall equation (3).

$$i_4 = \frac{i_3 + 14}{4}$$

Substitute 2 for i_3 .

$$i_4 = \frac{2 + 14}{4}$$

$$= \frac{16}{4}$$

$$= 4 \text{ A}$$

Step 8 of 11

Recall equation (2).

$$i_2 = 1 + i_3$$

Substitute 2 for i_3 .

$$i_2 = 1 + 2$$

$$= 3 \text{ A}$$

Step 9 of 11

Calculate the current I_1 .

$$I_1 = i_2 - i_1$$

Substitute 4 for i_1 , and 3 for i_2 .

$$\begin{aligned} I_1 &= 3 - 4 \\ &= -1 \text{ A} \end{aligned}$$

Therefore, the value of current I_1 is -1 A.

Step 10 of 11

Calculate the current I_2 .

$$I_2 = i_4 - i_1$$

Substitute 4 for i_1 , and 4 for i_4 .

$$\begin{aligned} I_2 &= 4 - 4 \\ &= 0 \text{ A} \end{aligned}$$

Therefore, the value of current I_2 is 0 A.

Step 11 of 11

Calculate the current I_3 .

$$I_3 = i_4 - i_3$$

Substitute 4 for i_4 , and 2 for i_3 .

$$\begin{aligned} I_3 &= 4 - 2 \\ &= 2 \text{ A} \end{aligned}$$

Therefore, the value of current I_3 is 2 A.